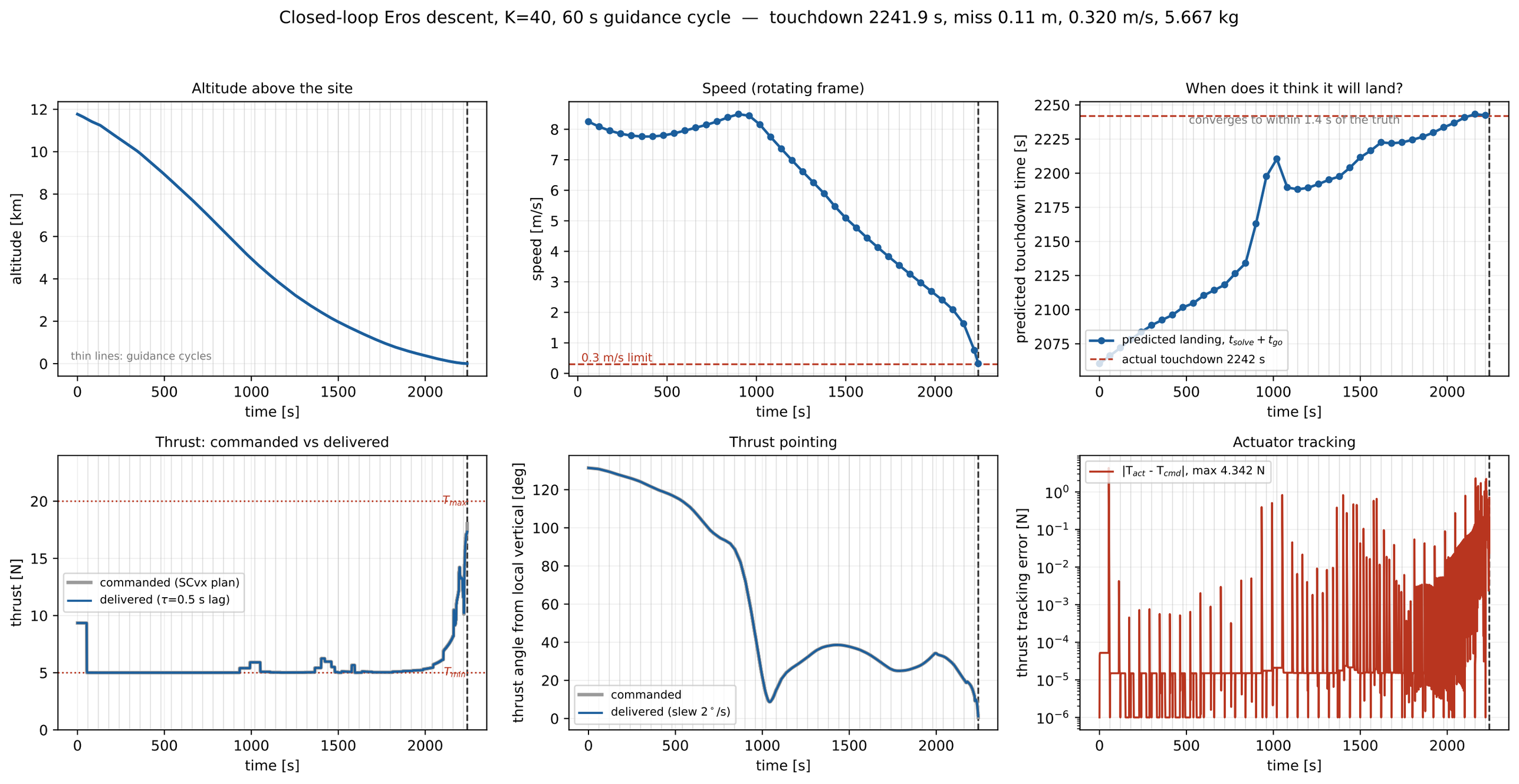

Closed-loop Eros descent, K=40, 60 s guidance cycle — touchdown 2241.9 s, miss 0.11 m, 0.320 m/s, 5.667 kg
Altitude above the site
altitude [km]
thin lines: guidance cycles
time [s]
Speed (rotating frame)
speed [m/s]
0.3 m/s limit
time [s]
When does it think it will land?
predicted touchdown time [s]
converges to within 1.4 s of the truth
predicted landing, $t_{solve} + t_{go}$
actual touchdown 2242 s
time [s]
Thrust: commanded vs delivered
thrust [N]
T_{max}
T_{min}
commanded (SCvx plan)
delivered (τ=0.5 s lag)
time [s]
Thrust pointing
thrust angle from local vertical [deg]
commanded
delivered (slew 2 °/s)
time [s]
Actuator tracking
thrust tracking error [N]
$|T_{act} - T_{cmd}|$, max 4.342 N
time [s]